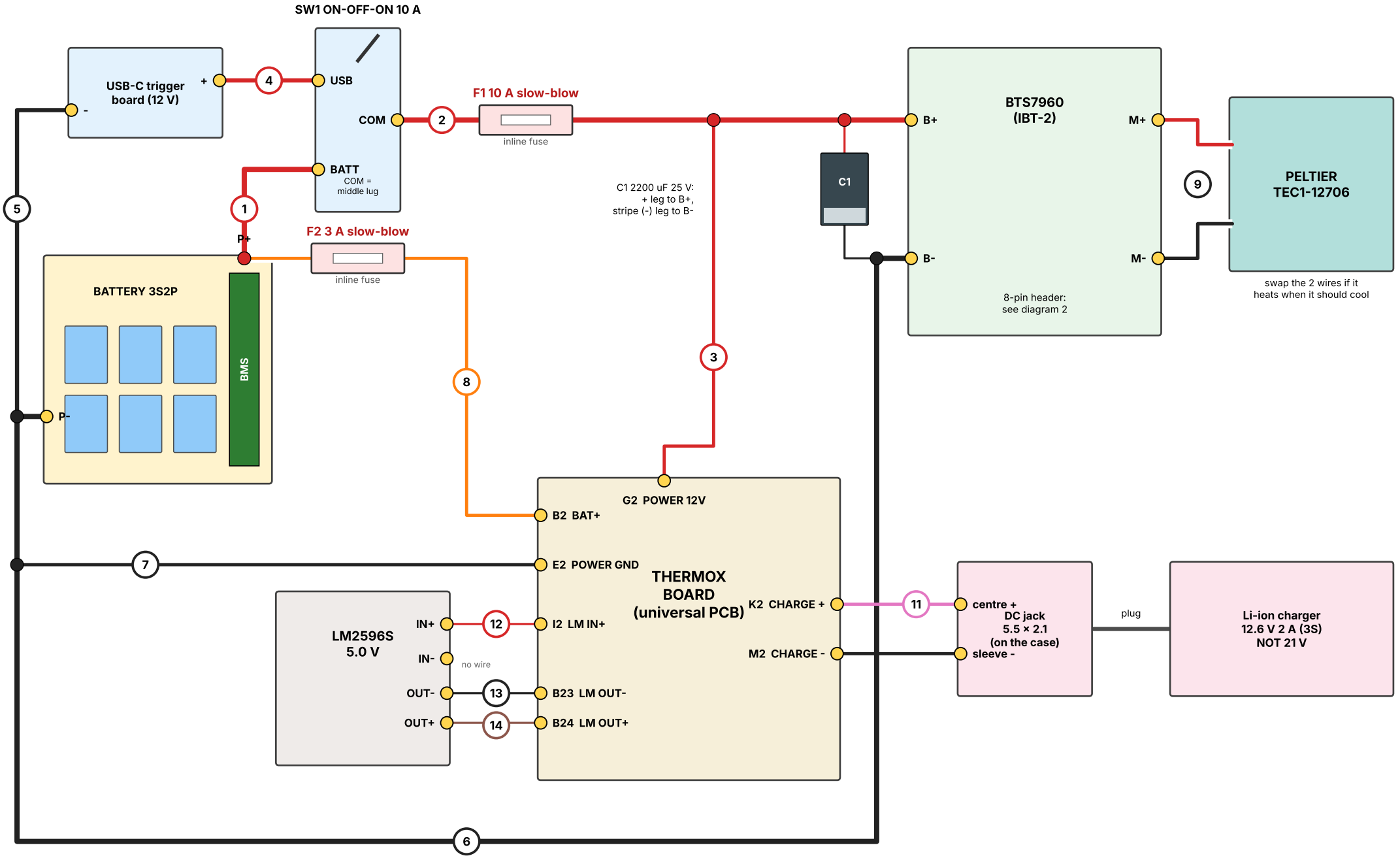
ThermoX WIRING DIAGRAM 1 - POWER [universal PCB]

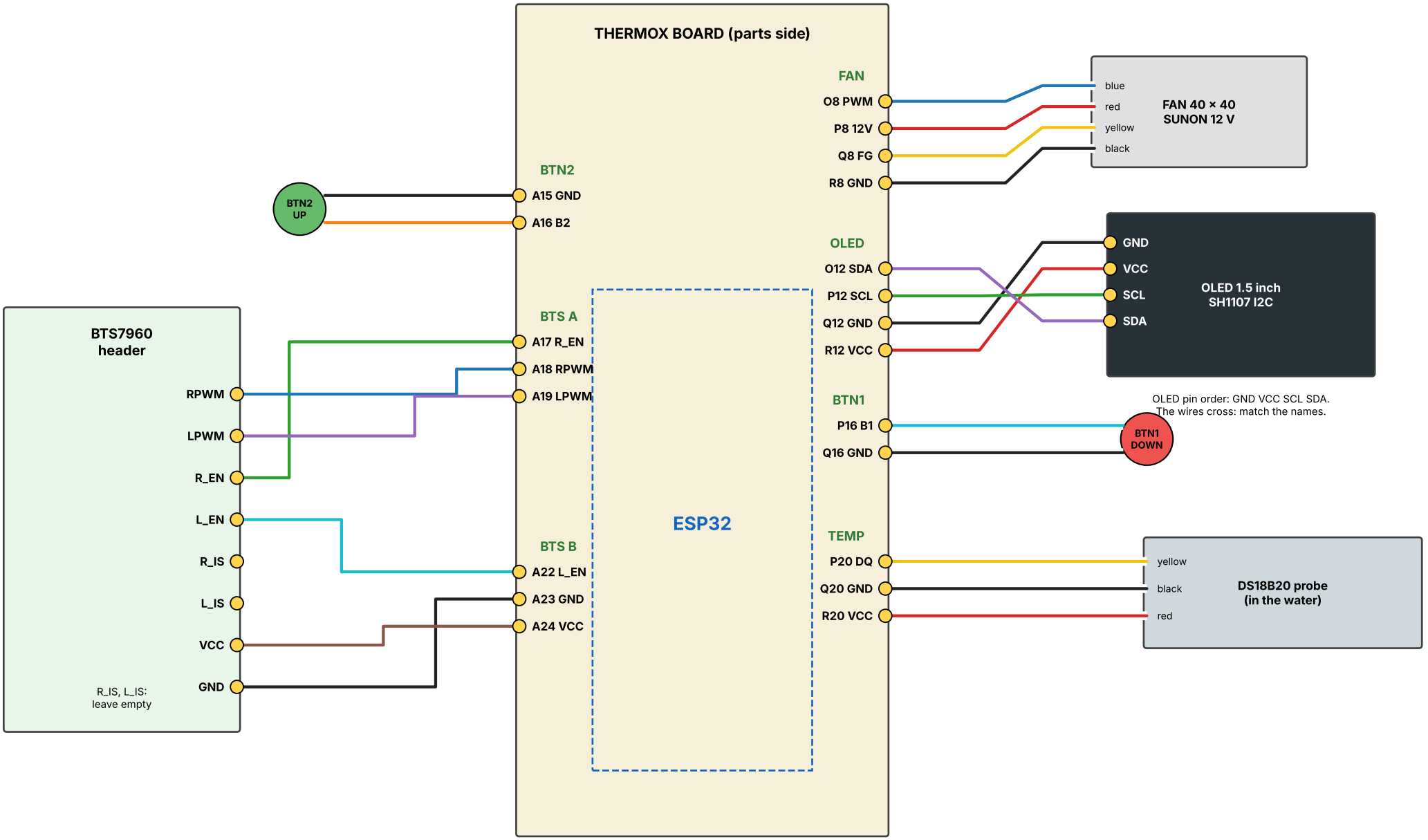
Thick lines = 1.5 mm2 (Peltier current). Dots = joined. Numbers = wire list on the next page. Lines that cross without a dot are NOT joined.



Before the battery goes on: SW1 in the middle (OFF), fuses in, LM2596 set to 5.0 V, trigger board set to 12 V. Never plug the USB-C charger and the DC charger in at the same time.

ThermoX WIRING DIAGRAM 2 - HEADERS TO THE PARTS (Dupont wires) [universal PCB]

Each line is one female-female Dupont wire. Connect by NAME: the order of the pins on a module can be different from the board.



The BTS7960 B+, B-, M+ and M- screw terminals are on diagram 1 (thick wires). Short Dupont wires are best: a loose plug can stop the fan or the driver.

ThermoX WIRE LIST [universal PCB]

Tick each wire when it is done.

POWER WIRES (diagram 1)

No.	From	To	Wire	Colour	Note
[ ] 1	Battery P+ (BMS)	SW1 BATT lug	1.5 mm2	red	
[ ] 2	SW1 COM (middle lug)	F1 10 A → BTS7960 B+	1.5 mm2	red	F1 inline in this wire
[ ] 3	Wire 2 after F1 (splice)	Board POWER 12V (G2)	0.75 mm2	red	solder + heat-shrink the splice
[ ] 4	Trigger board +	SW1 USB lug	1 mm2	red	
[ ] 5	Trigger board -	Battery P- (BMS)	1 mm2	black	Peltier current on USB-C
[ ] 6	Battery P- (BMS)	BTS7960 B-	1.5 mm2	black	
[ ] 7	Battery P- (BMS)	Board POWER GND (E2)	0.75 mm2	black	
[ ] 8	Battery P+ (BMS)	F2 3 A → Board BAT+ (B2)	0.75 mm2	orange	F2 within 5 cm of P+
[ ] 9	BTS7960 M+ and M-	Peltier red and black	Peltier wires	red/black	swap them if it heats when it should cool
[ ] 10	C1 2200 uF + leg / stripe (-) leg	BTS7960 B+ / B-	C1 legs	-	in the screws with wires 2 and 6
[ ] 11	DC jack centre (+) / sleeve (-)	Board CHARGE + (K2) / - (M2)	0.5-0.75 mm2	pink/black	
[ ] 12	LM2596 IN+	Board LM2596 IN+ (I2)	0.5 mm2	red	
[ ] 13	LM2596 OUT-	Board B23	0.5 mm2	black	LM2596 IN- needs no wire (same as OUT-)
[ ] 14	LM2596 OUT+	Board B24 (5 V)	0.5 mm2	brown	set the LM2596 to 5.0 V first

DUPONT WIRES (diagram 2)

```
[ ] FAN    O8   PWM   -> fan BLUE wire
[ ] FAN    P8   12V   -> fan RED wire
[ ] FAN    Q8   FG    -> fan YELLOW wire
[ ] FAN    R8   GND   -> fan BLACK wire
[ ] OLED   O12  SDA   -> OLED SDA
[ ] OLED   P12  SCL   -> OLED SCL
[ ] OLED   Q12  GND   -> OLED GND
```

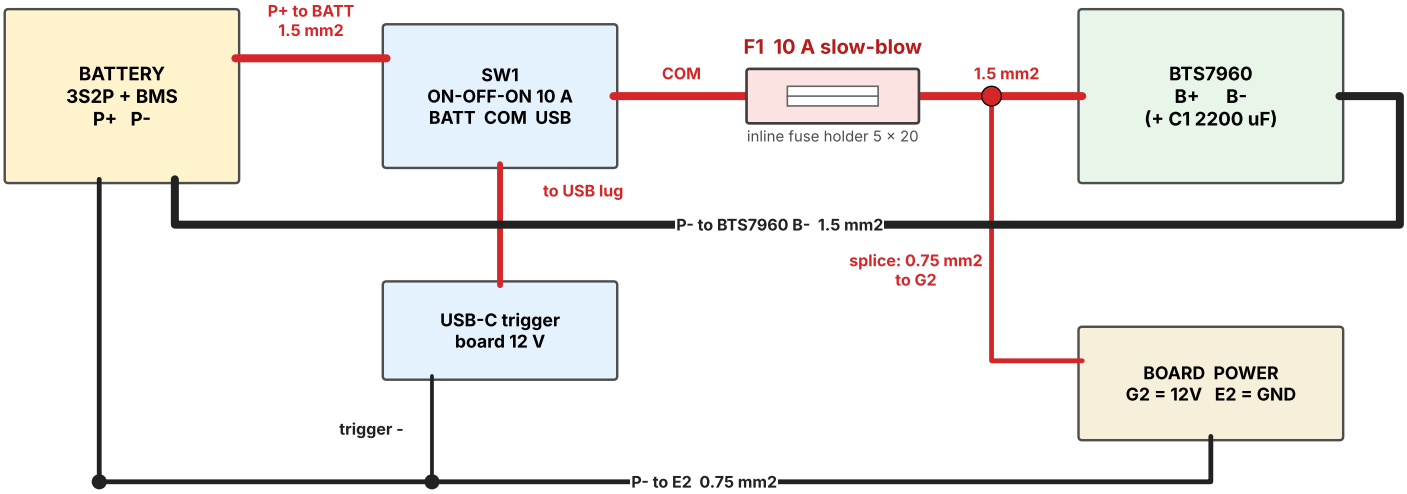
```
[ ] OLED   R12  VCC   -> OLED VCC (3.3 V)
[ ] BTN1   P16  B1    -> button 1 (DOWN), one leg
[ ] BTN1   Q16  GND   -> button 1, other leg
[ ] TEMP   P20  DQ    -> DS18B20 YELLOW (data)
[ ] TEMP   Q20  GND   -> DS18B20 BLACK
[ ] TEMP   R20  VCC   -> DS18B20 RED
[ ] BTN2   A15  GND   -> button 2, other leg
```

```
[ ] BTN2   A16  B2    -> button 2 (UP), one leg
[ ] BTS A  A17  R_EN  -> BTS7960 R_EN
[ ] BTS A  A18  RPWM  -> BTS7960 RPWM
[ ] BTS A  A19  LPWM  -> BTS7960 LPWM
[ ] BTS B  A22  L_EN  -> BTS7960 L_EN
[ ] BTS B  A23  GND   -> BTS7960 GND
[ ] BTS B  A24  VCC   -> BTS7960 VCC (5 V)
```

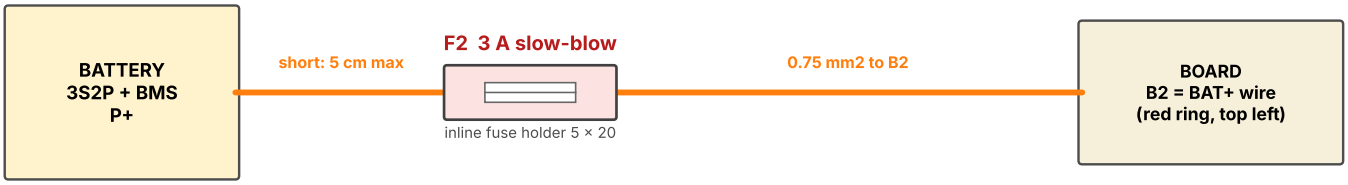
ThermoX - 9. FUSES AND FAN WIRES [universal PCB]

The fuses are NOT on the board on purpose: the Peltier current (up to 7 A) must not go through a perfboard.

1. F1 10 A - main 12 V line (Peltier, fan, board)



2. F2 3 A - BAT+ wire to the board (battery gauge and charging)



HOW TO FIT EACH INLINE FUSE HOLDER

1. Battery disconnected (or BMS off) while you wire. SW1 in the middle (OFF).
2. Cut the wire where the fuse goes. Strip 8 mm of each end and of both holder leads.
3. Slide heat-shrink on first, then twist each end to one holder lead, solder, and shrink the tube over the joint.
4. F1: the splice to G2 is on the BTS7960 side of F1 (after the fuse). Solder the 0.75 mm2 wire to the 1.5 mm2 wire and cover it with heat-shrink. Both the BTS7960 and the board are then protected by F1.
5. F2: put it within 5 cm of the battery P+ so the whole thin wire to B2 is protected. Solder the F2 lead and the 1.5 mm2 SW1 wire both on the BMS P+ pad.
6. Put the fuses in: 10 A slow-blow in F1, 3 A slow-blow in F2. Never a bigger fuse, never a wire instead.
7. Check with a meter (fuses in, battery off): SW1 COM to G2 beeps (through F1); battery P+ to B2 beeps (through F2).
8. A fuse that blows means a short: find it before you put a new fuse in.

FAN: SUNON 40 × 40 × 28 mm, 12 V, 4 wires → FAN header O8-R8 (row 8, lies across)

BLUE → O8 PWM: speed control, GPIO23	**RED → P8** 12V: +12 V (after F1)	**YELLOW → Q8** FG: speed signal (rpm), GPIO35	**BLACK → R8** GND: ground

These are the usual SUNON 4-wire colours. Check the sticker or datasheet of YOUR fan first: some fans use other colours. If the fan only has 3 wires (no blue), it has no PWM: leave O8 empty, it runs full speed whenever 12 V is on. With only 2 wires there is no FG either: set FAN_MIN_RPM = 0 in ThermoX.ino (stall check off).
The fan needs about 1 A at full speed. Use the Dupont cable only if it is short; a loose plug makes the fan stall.